

Beyond Current Initiatives: ITMO Regulations to Ratchet up Ambition

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Abstract

The various principles, obligations, and partnerships forming the international climate policy regime are insufficient to achieve the Paris Agreement temperature target. Although Internationally Transferred Mitigation Outcomes (ITMOs) aim to make mitigation more efficient, their unfettered use could weaken the domestic ambition of countries buying them. To bolster ambition and align Nationally Determined Contributions (NDCs) with the Paris target, this paper proposes that a coalition of willing countries commit to additional rules governing ITMO use and NDC definition. Basic rules include the harmonization of NDC targets using a comparable metric and requiring that no ITMO be purchased from a country missing its NDC. Then, additional rules with increasing degrees of ambition are described. First, restrictions based on national assessments are discussed, in which coalition members commit not to buy ITMOs from countries with unambitious NDCs. Next, it is argued that NDCs should be assessed for the coalition as a whole rather than country-by-country. Participating countries would cooperate in determining their NDCs to ensure joint alignment with the Paris target and commit not to exchange ITMOs with countries outside the coalition. Then, the study provides a comparative assessment of alternative proposals intended to reinforce the international climate policy regime, including differentiated carbon price floors and a supply-side policy known as *fossil-fuel non-proliferation*, and argues they lack the necessary political realism or efficiency. The article concludes that combining a tightly regulated ITMO coalition with ambitiously expanded Just Energy Transition Partnerships offers a viable pathway for effective international decarbonization.

Keywords: Climate policy; carbon price; ITMOs; carbon trading.

JEL: Q56; F38; H23; Q54; H87; F64; Q58; F53; F35.

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Policy Insights

- Unfettered use of ITMOs risks weakening decarbonization efforts of buyer countries, unless the global sum of NDCs align with the Paris temperature target.
- As a prerequisite to stricter rules concerning ITMOs, NDCs should be harmonized: they should cover all gases, be defined in absolute terms, and ideally include a cumulative emissions target.
- Countries should not buy ITMOs from a country which is failing its NDC target.
- A coalition of countries could submit a joint NDC in line with the Paris target and commit not to exchange ITMOs with countries outside the coalition.
- Alternatives to carbon trading are often inefficient, unfair, or unrealistic. For example, fossil fuel non-proliferation requires the unlikely cooperation of fossil fuel producers.

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Introduction

In 2015, governments universally adopted a global temperature target at the Paris Agreement. While countries have developed various commitments, mechanisms, and initiatives to stabilize the climate, they are still insufficient to achieve the target. This paper reviews the international climate policy regime, assesses existing proposals to ratchet up ambition, and delineates new rules for a coalition of countries to institutionalize higher ambition within cross-border carbon trading.

At COP29 in 2024, the international community finalized the rules for operationalizing Article 6 of the Paris Agreement, which allows cross-border carbon trading. In the core of this paper, I discuss the consequences of Article 6.2, which allows carbon trading at the intergovernmental level in the form of Internationally Transferred Mitigation Outcomes (ITMOs). While the Article 6 is intended to foster higher ambition by directing mitigation funding where it is most effective, there is a risk that the unfettered use of ITMOs could result in the weakening of climate action. Indeed, some developing countries may have specified unambitious NDCs in the hope of selling more ITMOs, whereas some developed countries that have originally considered an ambitious domestic mitigation pathway may later decide to purchase emission reductions abroad as ITMOs offer a cheaper alternative to domestic action. To the extent that NDCs remain collectively inconsistent with the Paris temperature target, ITMOs would be underpriced (unless their use is restricted) since they would not reflect the constraint implied by the Paris target. While authors such as ? have proposed restrictions on the use of ITMOs to ensure that they do not undermine existing ambition, I propose more stringent rules to strengthen ambition and align carbon trading with the Paris Agreement temperature target.

On a related topic, ? offered recommendations to strengthen the integrity of carbon offset markets. They argued for assessing a project's emission reduction relative to a *dynamic baseline* with counterfactual emissions converging to zero at a date depending on the country's development level, rather than relative to a business-as-usual scenario incompatible with the Paris target. This proposal eventually inspired certain rules operationalizing Article 6.4 of the Paris Agreement (?). With this article, I contribute to the parallel debate on how Article 6.2 can be used to ratchet up ambition. In particular, I make the case for a coalition of the willing to jointly align its NDCs with the Paris target and commit not to exchange ITMOs with countries outside the coalition.

The paper is divided into three largely independent sections. Section 1 contextualizes the paper through a critical assessment of the current international climate policy regime, with particular attention to ITMOs. In Section 2, I discuss various proposals for aligning ITMOs with the Paris temperature target. I start by describing existing proposals by ? and ?. Then, in the core contribution of the paper, I propose different options for leveraging ITMOs to enhance climate ambition, with increasing levels of stringency. Table 1 summarizes these proposals. Finally, in Section 3, I discuss alternative proposals to strengthen climate action, from differentiated carbon price floors to a fossil fuel non-proliferation treaty, and show that they lack efficiency or realism compared to the proposed regulations of ITMOs. Section 4 concludes. In the Supplementary Material, I provide three

tables summarizing the evaluation of each policy mentioned in this article.

1 A critical assessment of the current regime

The international climate policy regime is established by the United Nations Framework Convention on Climate Change (UNFCCC) and the Paris Agreement adopted under it, which sets a global temperature target. The global consensus supporting this regime is an immense success: the UNFCCC has been universally adopted, and the Paris Agreement had been ratified by all countries but three (Iran, Libya, and Yemen) before the U.S. withdrawal. However, reliance on consensus for decision-making within the UNFCCC results in major limitations: agreements reflect the lowest common denominator and fall short of driving any substantial progress on international climate action. In this section, I review the current regime and its most likely developments.

1.1 Burden-sharing

In 1992, the UNFCCC introduced a distinction between developed and developing nations: the former shall provide financial resources to the latter to promote their sustainable development and climate action. This classification is now outdated, stalling progress in critical negotiations, as newly high-income countries resist being reclassified as developed, and historically developed countries are reluctant to increase their contributions unless all high-income countries do so (???).

While high-income countries are obligated to provide resources to foster climate action in lower-income countries, the determination of required transfers would be more appropriately based on up-to-date, continuous indicators such as GNI per capita, rather than an outdated binary classification. A simple yet fair rule would be that a country's contributions should be made in proportion to GNI and entitlements in proportion to population (?).

In contrast, the UNFCCC only refers to the vague principle that protecting the climate should be based on "Common But Differentiated Responsibilities" (CBDR). As the key issue of the burden-sharing rule was left unresolved by the CBDR principle and its various possible interpretations, countries have not been able to agree on binding targets for emission reductions and financial transfers between countries. Consequently, NDCs are collectively insufficient, projecting 2.6 °C of warming according to [Climate Action Tracker](#). Meeting the 2 °C target with 67% probability requires global emissions to drop to 41 GtCO₂e by 2030 (?), far below the 51 Gt projected under current NDCs.

1.2 Climate finance

Substantial international transfers are required to achieve the Paris temperature target under a fair allocation of the remaining carbon budget. An equal per capita allocation of emissions rights would entail transfers from high to low emitters of \$1.5 trillion in 2035

and of over \$1 trillion annually until 2060.¹ Taking historical responsibility for emissions into account, an equal per capita allocation of cumulative (past and future) emissions rights would entail even larger transfers; the “carbon debt” that the North owes to the South is estimated at \$26 to \$192 trillion (??).

At COP29, the international community reached a compromise on the New Collective Quantified Goal (NCQG). Developed countries committed to mobilizing \$300 billion per year by 2035 for climate action in developing countries. However, the NCQG can be reached through loans (including from the private sector), and does not specify what share should be provided as grants (or grant-equivalent concessional loans). In fact, the current goal of \$100 billion is met with only \$26 billion provided as grants (?).

In contrast, at COP29, India called for at least \$600 billion in the form of grants and equivalent resources (?). Global South countries stress that an insufficient provision of climate finance jeopardizes their decarbonization, since many of them condition their NDC on the adequate provision of climate finance.

Together with increased North-to-South transfers, reforms to the international financial system are needed to reorient resources towards climate action, such as debt relief for low-income countries, a new issuance of at least \$650 billion in Special Drawing Rights by the IMF to expand loans of Multilateral Development Banks (MDBs) to at least \$500 billion per year, and public guarantees to lower interest rates on sustainable projects in the Global South (?).

While scaling up climate finance is crucial, it is not sufficient to decarbonize the world, as it does not cap (or directly reduce) emissions. In the worst-case scenario, the expansion of low-emission projects would merely add low-carbon infrastructures on top of fossil ones, failing to meaningfully reduce emissions.

1.3 Just Energy Transition Partnerships

Just Energy Transition Partnerships (JETPs) are agreements where a developing country essentially commits to emission reductions through the deployment of renewable energy in exchange for concessional terms on the required loans provided by a group of developed countries. Four JETPs have been signed to date, involving Indonesia, Vietnam, South Africa, and Senegal, with \$2 to \$20 billion in pledged loans (?).

While JETPs offer a promising way to deliver climate finance that guarantees emission reductions, they currently suffer from several shortcomings. First, their coverage is limited regarding sectors and countries. To improve sectoral coverage and efficiency of JETPs, researchers have proposed designing them as financial transfers in exchange for a national carbon price (?). Second, as they focus on emission reductions rather than sustainable development, JETPs do not contribute to poverty reduction. This concern could be mitigated by designing JETPs with a higher reliance on grants (?). However, increasing the provision of grants is challenging absent a dedicated source of revenue (such as an international tax). Lastly, even if JETPs were improved accordingly, they would still

¹I computed these figures using the model NICE under a 1.8°C scenario.

fail to guarantee that the decarbonization of major emitters like China or the European Union is consistent with necessary global efforts.

1.4 Internationally Transferred Mitigation Outcomes

Article 6.2 of the Paris Agreement allows Parties to exchange Internationally Transferred Mitigation Outcomes (ITMOs). This enables a country to nominally reduce its emissions (for the purpose of NDC assessment) by purchasing verified emission reductions from another country. The purchased ITMOs are then added to the emissions account of the seller country through a corresponding adjustment.

As any bilateral agreement on ITMO is permitted, the use of ITMOs risks reducing buyers' domestic decarbonization efforts. Indeed, to the extent that the NDCs do not add up to the global emission reduction objective, there will be some "ambition gap": ITMOs will not reflect the required mitigation constraint, and their price will be too low. As a result, ITMOs may propagate a global lack of ambition to countries with otherwise ambitious NDCs, offering a cheap (and less effective) alternative to domestic decarbonization.

To illustrate this, consider a fictive example with two world regions, Rich and Poor. Assume the global carbon budget is 1,000 Gt and that in a business-as-usual scenario without climate action, both regions would emit 750 Gt. Imagine that region Rich sets an ambitious NDC of 500 Gt, while Poor sets a low ambition NDC of 1,000 Gt. In the absence of international carbon trading, region Rich would be expected to emit 500 Gt (in line with its NDC) and region Poor to emit 750 Gt (as no climate action is required to fulfill its NDC). In this example, region Poor may be willing to sell 250 Gt of ITMOs to region Rich at a very low price. Region Rich could then meet its NDCs while emitting 750 Gt, resulting in global emissions of 1,500 Gt, higher than the 1,250 Gt that would have occurred without ITMOs.

According to Climate Action Tracker, the sum of current NDCs would lead to global warming of around 2.6°C, far above the Paris target of "well below 2°C." In such a context, international carbon trading risks becoming a vehicle for exporting low ambition rather than reinforcing collective effort. Buyers will find it more attractive to purchase credits abroad than to accelerate the domestic transformation of their economies. In short, unregulated ITMOs can result in "emissions dumping".

This dynamic is already apparent. The European Union now allows the use of international credits toward its 2040 climate target. More precisely, up to 5% of 1990-level emissions could be offset through ITMOs (?). Since the European Commission aims to cut EU emissions by 90% by 2040, relying on ITMOs could enable the EU to emit 50% above its 2040 domestic goal through purchasing emission reductions abroad. Likewise, Japan plans to emit up to 53% above its goal in 2040 (?). Switzerland has already purchased ITMOs from Thailand and Ghana and may rely on ITMOs for over 50% of its 2035 goal. Adding intended purchases from South Korea, Norway, and Singapore, the market for ITMOs could reach 500 MtCO₂ per year by 2040.

Meanwhile, China pledges to reduce emissions by 7–10% from their peak by 2035 (?).

Yet, a reduction of around 20% would be needed to align with a 2°C scenario (?). If China or other countries were to sell ITMOs based on such unambitious trajectories, buyers would acquire credits that do not represent genuine progress toward the Paris goal.

To prevent ITMOs from weakening domestic action, countries that use them should commit to extra rules that go beyond merely verifying the environmental integrity of the credits.

2 Aligning Carbon Trading with the Paris Target

The present section reviews existing proposals to regulate ITMOs and considers new ones. All proposed analyzed in this section are summarized in Table 1.

2.1 Existing proposals

? propose verifying that ITMOs are additional, meaning they correspond to emission reductions relative to a counterfactual scenario without such ITMOs. The authors suggest a two-step procedure wherein a new “Article 6 Supervisory Board” would first check whether the seller’s NDC is more ambitious than the business-as-usual (BAU) trend. If so, the second step would be unnecessary. Otherwise, the seller’s unambitious NDC generates “hot air”, requiring ITMO to be tested at the project level. In this second step, the additionality of each specific activity financed by the ITMO would be assessed individually.

? discuss restricting ITMOs to emission reductions below the BAU. They compute BAU emissions for a range of countries using four potential definitions for the BAU: extending past emissions levels, extending emission intensity, or projecting their respective historical trend. They show that none of these definitions is satisfactory, as each would still generate *hot air* for some countries where the BAU exceeds emissions as projected by Climate Action Tracker. The authors propose instead to limit the quantity of ITMOs that a country can sell to a fixed share of its emissions (e.g. 1% or 5%). They acknowledge that quantity limits would still allow some hot air but argue that such limits can be calibrated to strike a balance between reducing hot air and exploiting the gains from trade.

While ? show that simple definitions of the BAU are ill-suited for regulating ITMOs, ? propose delegating the determination of the BAU to independent experts so that NDC ambition can be accurately assessed.² In both papers, the authors agree that an NDC can be assessed at the national level, by comparing it to the country’s projected emissions, and that ITMOs should be allowed whenever the NDC is more ambitious than properly projected emissions.

²Expert projections of BAU emissions already exist, see e.g. ??.

2.2 Desirable paths to regulate carbon trading

Principled buyers of ITMOs could also employ other national criteria to assess the adequacy of a seller's NDC. Here are different original proposals, ordered by degree of stringency.

A necessary precondition: harmonized NDCs. Currently, few constraints apply to the definition of NDCs. Countries independently choose their sectoral and gas coverage, the methods for converting different gases into CO₂-equivalent, and whether to use an absolute or carbon intensity target, etc. As a result, NDCs are not harmonized. Researchers must make assumptions (e.g. on GDP growth or LULUCF emissions) to express them using a comparable metric, and estimates of NDC targets vary significantly across research teams.³

A necessary precondition to assessing NDCs is the strengthening of reporting requirement. NDCs should cover all Kyoto gases and be expressed in absolute terms using harmonized conversion factors between gases (namely GWP100, the most up-to-date being from the IPCC AR6). Furthermore, they should encompass all sectors—broken down individually, or at least separating LULUCF from non-LULUCF emissions. Ideally, NDCs should also specify a country's future cumulative emissions, its planned emission trajectory, and the intended use of ITMOs. These enhanced reporting requirements would allow for the proper assessment of NDCs and enable conditioning the use of ITMOs on the achievement and adequacy of NDCs.

A sine qua non condition: no ITMOs from failed NDCs. As a minimal requirement, countries should be allowed to sell ITMOs only up to the amount by which their emissions fall below their NDC targets. Consequently, buyers should refrain from purchasing ITMOs from a country that is failing to meet its NDC target. This principle should also apply to countries that have weakened their emission targets (instead of ratcheting up ambition), resulting in emissions higher than planned under a previous target.⁴

Principled buyers: no ITMOs from unambitious NDCs. Furthermore, to strengthen the additionality requirement proposed by ?, principled buyers should refuse to buy ITMOs from a country whose NDC target is above the BAU or evidently incompatible with the Paris target, even when it is below the BAU. Such incompatibility could be safely assumed when a country's NDC target exceeds both its cost-optimal emissions required to limit global warming to 2°C and its equal per capita share of global emissions under that 2°C scenario. Indeed, no credible burden-sharing rule could then justify the NDC's adequacy with the Paris temperature target. Given that there are multiple ways to define a

³Estimates of emissions implied by NDC targets are provided by [Climate Action Tracker](#), [UNDP](#), [Climate Analytics](#), [Climate Resource](#), [PBL](#), [Climate Watch](#), ??.

⁴Angola provides an example of a country that effectively decreased its ambition in an updated NDC, by revising upwards (by a factor two) its BAU emissions.

cost-optimal 2°C scenario (as the carbon budget depends on the probability of achieving the 2°C target, while the trajectory depends on the model used), simple substitutes could be employed instead. In particular, buyers could refuse to acquire ITMOs from countries with per capita emissions above the world average.

The first exchange of ITMO that has been finalized was between Thailand and Switzerland. Thailand appears to meet all aforementioned criteria, suggesting that Switzerland could be considered a principled buyer. However, until the sum of all countries' NDCs aligns with the Paris target, the price of ITMOs might be too low. In other words, the above requirements are necessary but not sufficient conditions to close the ambition gap.

Adequate climate finance unlocking conditional NDCs. While these requirements would reduce the amount of hot air, they would exclude from the trade of ITMOs low-emitting countries that set their NDC target above the BAU,⁵ at (what they consider to be) their fair share. These potential sellers of ITMOs could join forces and use their market power to negotiate guarantees of grant-based climate finance at scale. In particular, these countries could commit to setting their NDC target below the BAU (and selling ITMOs) in exchange for a commitment from buyers to fairly allocate taxing rights of new international levies, or to finance JETPs or debt relief.

There is another way sellers would incentivize principled buyers to provide sufficient climate finance. Until now, I conflated conditional and unconditional NDC targets. Yet, Global South countries often include both types of target: the conditional target sets more stringent emission reductions than the unconditional one, but applies only at the condition that the country receives a specified amount of climate finance from developed countries. Let me call *unidirectional climate finance* all climate finance flows that do not rely on ITMOs. When assessing a country's NDC (to determine whether ITMOs can be purchased from that country), the conditional NDC target should replace the unconditional target in the adequacy assessment once the country's conditions (in terms of climate finance) are met, counting only unidirectional climate finance. Therefore, if a country sets its unconditional target above the BAU and its conditional target below the BAU, principled buyers would have to provide climate finance to that country in order to purchase ITMOs from it, otherwise the seller's NDC would not be ambitious enough.

2.3 Limitations of the previous options

Issues with national assessments of NDCs. The aforementioned proposals rely on assessing NDCs against national BAUs. However, this approach raises two issues. First, even if a country's NDC target is set below its BAU, to the extent that NDCs do not align with the Paris temperature target when aggregated at the global level, the ambition gap will persist. Consequently, both the demand for ITMOs and their price will remain too low. Second, given that grant-based climate finance falls short of their demands, Global South countries could legitimately set up unconditional NDC emission targets above their

⁵For example, ? conclude that India has set its 2030 NDC above the BAU.

BAU emissions to use extra emission space to sell ITMOs. Actually, insofar as NDC targets represent how countries should share the burden of emission reductions, low-income countries already claim less than their fair share of emissions.⁶ Therefore, the ratcheting up of ambition should arguably be borne by industrialized countries. For these reasons, assessing an NDC against the national (or project-based) BAU is neither a fair nor effective way to close the ambition gap.

A rule restricting the sale of ITMOs to cases where the NDC target is below the BAU would be satisfactory only at a global scale or within a large coalition, since this is the scale at which there is consensus that the BAU is inadequate. Conversely, at the national level, some countries may have already implemented stringent climate policies such that their BAU emissions are ambitious enough, while others would legitimately set an NDC target above their BAU since they deserve to sell ITMOs. In both cases, a national comparison of the NDC to the BAU would be inappropriate.

The key problem with ITMOs is that the *global* emissions implied by current NDCs (or current trends) do not align with the Paris temperature target. Meanwhile, given the disagreement over what constitutes a fair share of the global carbon budget, countries cannot agree on whether a given NDC complies with the Paris target or claims an excessive carbon budget. However, if a critical mass of countries agreed on a common norm, such as a burden-sharing principle or on a decision rule to assess whether the global carbon budget is fairly shared, then the two issues of effectiveness and fairness could be resolved.

Limitations of restrictions to the use of ITMOs. As ? explain, restricting the use of ITMOs involves a trade-off between limiting hot air and exploiting gains from trade. Furthermore, the restrictions discussed above present additional limitations. They rely on arbitrary cutoffs, which create undesirable discontinuities in the ability to sell ITMOs. Additionally, they do not provide an institution for states to negotiate common ground regarding overall ambition or the scale of climate finance.

A coalition of willing countries could establish precise norms for NDC adequacy and climate finance contributions. The adoption of such norms by a large group of countries would alleviate the need for restrictions on ITMO exchanges between them (beyond the *sine qua non* condition) and overcome the aforementioned limitations.

2.4 The case for a joint definition of NDCs

A coalition of the willing could even go further and submit a joint NDC (just as the European Union does). The common norm would be used to verify that the joint NDC complies with the global target and thereby closes the ambition gap. This coalition could also allocate its aggregate NDC between countries in a way deemed acceptable and fair, potentially allowing certain countries to get a target higher than their BAU emissions.

⁶Based on my own estimates, the 2030 NDC targets of Global South countries (excluding China) total 13.5 GtCO₂, that is 2.8tCO₂ per capita, which is below equal per capita emissions aligned with the Paris target.

Table 1: Proposals to strengthen ambition using ITMOs or NDCs. Restrictions based on national assessments of NDCs are denoted with *Nat*, those based on BAU trajectory are denoted with *BAU*.

Policy	Type	Description
Harmonized NDCs		NDCs should cover all sectors and GHGs, use an absolute target, and specify cumulative emissions.
ITMOs should be additional (?)	Nat, BAU	ITMOs should either be sold by a country whose NDC target is below the BAU, or their additionality verified at the project level.
ITMOs should be limited (?)	Nat	ITMOs should not exceed a fixed share (e.g. 1% or 5%) of the seller's emissions.
No ITMOs from failed NDCs	Nat	ITMOs sold should not exceed the difference between the NDC target and actual domestic emissions.
No ITMOs from unambitious NDCs	Nat, BAU	ITMOs should not be bought from a seller with unambitious NDC target, i.e. a target above the BAU or both above cost-optimal and equal per capita 2°C emissions.
No ITMOs from large emitter	Nat	ITMOs should not be bought from a seller with emissions above the world average.
Ambitious NDC in exchange for adequate climate finance		Potential sellers would set their (conditional) NDC target below the BAU in exchange for adequate climate finance.
Conditional NDC prevailing in case of adequate climate finance		The adequacy of a seller's NDC (to sell ITMOs) would be assessed using the conditional NDC target instead of the unconditional one iff the climate finance it receives matches the condition specified in its NDC.
ITMO coalition with a joint NDC		A coalition of countries would submit a joint NDC aligned with the Paris target and exchange ITMOs only among themselves. The NDC would ideally specify a carbon budget and yearly national targets.

361 Ideally, the joint NDC would include a carbon budget aligned with the Paris target,
362 broken down into yearly national targets. That way, the adequacy of emission trajec-
363 tories could be verified jointly and country-by-country, and the system could easily evolve
364 into a fully-fledged compliance carbon market. Alternatively, the joint NDC would de-
365 fine a minimum emission reduction rate, aligned with the Paris target. Initially, this rate
366 could be expressed in terms of emission intensity, consistent with the practice of emerging
367 economies. For example, the coalition's GHG emissions relative to output or final energy
368 use would need to decrease by 2% each year. In the medium term, the reduction should
369 be defined in absolute terms.

370 The countries most likely to join such a coalition are those with moderate emissions.
371 Therefore, there is a tension between setting the coalition's carbon budget based on a cost-
372 optimal allocation of global emissions (favoring large emitters outside the coalition) or an
373 egalitarian allocation (which might not sufficiently strengthen decarbonization ambition).

As a compromise, the coalition's carbon budget could be defined as an equal per capita share of a global carbon budget achieving 2°C with a 75% probability. Examples of other definitions with similar levels of ambition include a cost-optimal share of a world carbon budget achieving 1.9°C with a 50% probability, or an equal per capita share of a global carbon budget achieving 1.8°C with a 50% probability.⁷ Negotiations within the coalition would be essential to defining fair shares.

2.5 A coalition for the use of ITMOs

Here is how negotiations could shape an agreement along the lines sketched above. A coalition of countries could agree to jointly define their NDCs and exchange ITMOs exclusively among themselves to ensure that their carbon trading does not undermine the ambition of the Paris Agreement. Once a large coalition of countries agree on the broad vision, this coalition could be taken as given. The coalition's emissions targets would be set below its projected emissions, with a gradually increasing wedge between BAU emissions and the target. Specifically, the 2026 target would correspond to the coalition's emissions, while the 2027 target would be slightly below the BAU, increasing the divergence in 2028 and beyond. This would realistically scale up the additional decarbonization effort over time until it reaches an emission reduction rate aligned with the Paris target. The resulting emission trajectory of the coalition should not exceed its equal per capita share of a global carbon budget achieving 2°C with a 75% probability, and would ideally be lower.

After determining the coalition's trajectory, the disaggregation into national targets would be negotiated. It would be useful to allocate national targets starting from a focal point, such as the minimum between the country's BAU trajectory, its NDC targets, and an equal per capita share of the coalition's emission trajectory. Extra emission rights could then be allocated to countries with the lowest incomes and to those experiencing the greatest welfare loss due to the increased decarbonization effort. The proposed focal point can be justified by the principle that, to strengthen ambition, a country's target should not be higher than projected, committed, or *fair-share* emissions, and by the norm that the equal right to emit per capita is the fairness benchmark.

If new countries join the coalition, they could propose a new allocation of emission targets. If the proposal is rejected by the coalition, the entire negotiation process would be restarted. The new allocation would then be adopted only if it were accepted by a majority within the coalition and provided it does not lead to reduced emission coverage (resulting from unsatisfied countries leaving the coalition) or to a lower projected ITMO price (which would indicate reduced ambition).

A scientific council would assist the coalition by determining BAU emissions, modelling the climate, economic, and distributive effects of the agreement, providing requested analyses, and proposing target allocations or suggesting other arbitration de-

⁷According to the model NICE by ? and extrapolating from the ?, for a climate coalition comprising China, the Global South, Europe, and Japan, these three carbon budgets would be close to 750 GtCO₂ over 2026–2100 (including LULUCF but excluding non-CO₂ gases).

cisions. Each participating country would be entitled to designate a team of scientists to represent them on the council, and these teams could overlap. In the event of disagreement within the scientific council, each team would hold voting rights proportional to the population of the country (or countries) that designated them.

2.6 Feasibility of the previous proposal

While the last proposal is the most ambitious analyzed here, I argue that it constitutes the most feasible pathway to guarantee emission reductions in line with the Paris target within a large coalition of countries.

First, relying on a coalition parallel to the UNFCCC would enable the achievement of ambitious agreements. Second, academic surveys show that majorities around the world would genuinely accept international climate policies, even knowing the cost to themselves (???). Third, the proposals presented in this article are grounded in the Paris Agreement (and in particular Article 6.2) and they would operationalize the long-standing CBDR principle. These proposals respond to a renewed interest in well-regulated international carbon markets, as evidenced by two initiatives launched at COP30: the *Open Coalition on Compliance Carbon Markets*, which already includes China, the EU, Brazil, and fifteen other countries; and the *Article 6 Ambition Alliance*, a set of ten countries including Switzerland, Norway, and Germany, which intend to finance or host some emission reduction activities without use of ITMOs towards their NDCs.

3 Comparison of alternative proposals for phasing out fossil fuels

In Section 1, I have reviewed the pros and cons of ITMOs, climate finance, and JETPS, which represent the international decarbonization initiatives with the greatest chance of implementation. In this section, I assess alternative proposals to guarantee global decarbonization, in particular the expansion of carbon pricing and the restriction of fossil fuel extraction.

3.1 Differentiated carbon price floors

Some authors propose that all countries in a coalition price carbon nationally, without cross-country revenue sharing but with differentiated carbon price floors based on income levels⁸ (??). In a prominent contribution, ? propose restricting the agreement to carbon-intensive manufacturing products (e.g. steel, aluminum, cement) and applying a carbon border adjustment mechanism (CBAM) only at the borders of the climate coalition (at \$75/tCO₂). This proposal would have the EU renounce its own CBAM in exchange

⁸Namely, the authors propose \$75/tCO₂ for high-income countries, \$50/t for upper middle-income countries, and \$25/t for lower-income countries.

for China pricing carbon emissions in its manufacturing sector — thereby increasing the price of its final products (?). This proposal would reduce global CO₂ emissions by 2%. It has little chances of being implemented, as China represents about 70% of emissions covered in the proposed coalition, and China appears unwilling to commit to the proposed carbon price level (\$50/t).

One may wonder why carbon prices should be differentiated across countries. From a theoretical perspective, absent any imperfections, a uniform carbon price coupled with cross-country transfers is optimal (?). Specifically, a uniform price is more efficient, and under a redistributive system, lower-income countries would gain purchasing power. Indeed, provided lower-income countries are allocated more emission rights than actual needs, they could in principle choose to keep their emissions stable while receiving a financial transfer. Yet, the high carbon price would provide incentives to decarbonize, allowing them to benefit from larger transfers.

Admittedly, four kinds of imperfections justify differentiated carbon prices across countries or sectors: divergent discount rates, the presence of country-specific distortive taxes, market power in trade, and constraints preventing cross-country transfers. Let us review them in turn. First, ? show that equalizing carbon prices across countries is inefficient “if the equilibrium features cross-country differences in discount rates (or interest rates)”, themselves due to inefficiencies in the allocation of capital.⁹ However, no model has been developed to know whether welfare would increase or decrease upon equalizing carbon prices (through emission trading). Second, ? explain theoretically and ? models numerically that existing distortions can be reduced by differentiating carbon prices across sectors or countries. For example, if aviation remains under-taxed, then a higher sector-specific carbon price improves welfare. Third, the terms-of-trade effect can be well understood taking the example of China, which possesses significant market power in the oil or manufactured goods markets. As an oil importer, China would benefit from a lower oil price. By taxing oil more than other sectors, China would reduce global demand for oil, thereby lowering its price and improving its terms-of-trade. Similarly, China has an interest in setting a higher carbon price on manufactured goods to increase the value of its exports. Fourth, differentiated prices are generally justified by the assertion that international transfers are politically infeasible (??). ? refer to an “efficiency-sovereignty” trade-off in climate policies: either carbon prices are differentiated and efficiency is lost, or a uniform price is applied alongside transfers to equalize efforts across countries. Because they deem transfers infeasible, they view differentiated prices as a necessary second-best solution.

However, there is a distributional equivalence between differentiated carbon prices and a uniform price with differentiated emission rights (?). More precisely, a uniform price combined with an appropriate allocation of emission rights can replicate the distribution of costs and benefits generated by any differentiated price-floor agreement, while typically delivering the same or greater global emission reductions. This observation

⁹While differentiated carbon prices might be optimal in a second-best world, the first best is in principle attainable through capital market reforms to correct their imperfections.

should prompt governments to question their apparent reluctance to implement transfers, given that they produce the same distributive effects as differentiated prices without associated inefficiency costs.

3.2 A uniform carbon price with international transfers

Several authors propose that a coalition of countries implement a uniform carbon price and share the revenue internationally (???????). While an equal per capita allocation is viewed as a fair and progressive way to share carbon pricing revenue (??), small departures from the equal per capita benchmark may incentivize more countries to join the coalition and prevent high-income countries from becoming net recipients of transfers (?).

A global cap-and-trade system would guarantee that decarbonization is aligned with climate objectives. Furthermore, it would enforce carbon pricing directly on the firms at the source of emissions, whereas the achievement of NDCs without a global price must rely on the goodwill of governments to develop comprehensive decarbonization plans.

While this proposal receives genuine support from majorities across countries (??), key governments may be reluctant to join such a coalition for fear of losing sovereignty. By proposing a regulation of carbon trading more closely aligned with the framework of the Paris Agreement, this paper seeks to reproduce the efficiency and fairness of a redistributive cap-and-trade while leaving countries free to implement the policies of their choice.

3.3 Country-level incentives to decarbonization

To allow countries flexibility in their implementation of carbon pricing, ? proposes a system of monetary rewards for countries based on the extent to which they price carbon above a benchmark or emit less than the average, with symmetric penalties for countries deviating from these thresholds in the other direction. While the increased flexibility makes this proposal more feasible compared to a uniform price, it has its own limitations. This system does not guarantee that countries will implement sufficiently ambitious decarbonization policies. By rewarding countries with high carbon prices and low emissions, this system may lead to transfers to wealthy countries (such as Norway). Finally, the system relies on the willingness of countries to pay penalties.

Proposals for a “refunding club” aim to restore incentive-compatibility in agreements that rely on international transfers, ensuring that countries are incentivized to pay their dues and remain in the coalition (??). These proposals involve country-specific (and potentially negative) initial payments and period-by-period refunds proportional to countries’ mitigation spending. Initial payments can accommodate any burden-sharing allocation. ? estimate that an initial fund of \$2.6 trillion (that is, 0.3% of world GDP over ten years) is required to make the grand coalition stable and close half the gap between the non-cooperative and the social optima. Although this amount could in principle be raised

through various means, such as a 2% tax on billionaire wealth (?), persuading countries to make the initial payment could prove insurmountable.

3.4 Supply-side policies such as *fossil fuel non-proliferation*

The *Fossil Fuel Non-Proliferation Treaty* has emerged as a prominent campaign to phase out fossil fuels. The call for a treaty (which does not refer to a specific legal proposal) has been endorsed by over one million individuals, four thousands organizations (including Greenpeace and the Climate Action Network International), and 101 Nobel laureates. While the petition only alludes to a consensual call for a “binding plan to end the expansion of new coal, oil and gas projects and manage a global transition away from fossil fuels”; campaign briefings and related academic research sketch out a more detailed plan (????).

The campaign refers to a plan called the *Fair Shares Phase Out*, which involves setting country-specific end dates for fossil fuel extraction (?). This approach allows a later phase-out for countries with lower incomes or higher dependence on fossil fuel extraction. For example, the U.S. would be required to phase out oil extraction by 2031, Russia by 2037, Saudi Arabia by 2041, and Iraq by 2050.

This plan is problematic for three reasons. First, it requires the participation of all countries that export fossil fuels, yet these countries are the least likely to take action on climate change. Second, by restricting supply rather than demand for fossil fuels, this plan would increase fossil fuel rents instead of generating carbon price revenue. Therefore, despite the plan being framed as fair, it would probably widen inequality compared to demand-side policies, since (predominantly rich) owners of fossil fuel resources would benefit while the lack of carbon pricing revenue would make it difficult to compensate low-income consumers for higher fuel prices. Admittedly, the plan also calls for North-to-South transfers to address the negative distributive effects, but it fails to include a specific proposal on how to fund these transfers, how to allocate them, let alone an assessment of overall distributive effects. Third, the aforementioned extraction end dates would also result in an inefficient location of fossil fuel extraction (?), with e.g. cheap oil from Qatar being phased out 13 years before “dirty” oil from Venezuela.

An alternative policy would exhibit similar properties without the inefficiency problem: a producer carbon price. Under this policy, producer countries would price carbon at the wellhead and retain the revenue from carbon pricing (or most of them). Some argue that producer countries would accept a producer carbon price as a compromise if climate-ambitious countries were willing to penalise them for refusing to cooperate. To achieve this, climate-ambitious countries would need to commit to faster decarbonization and trade sanctions on fuel exporter countries (further reducing their revenues) if they fail to price carbon (?). Compared to an equal per capita allocation of carbon price revenues, this solution lacks equity as it grants tax revenues to producer countries, most of which are wealthy. Furthermore, proponents acknowledge that the proposal hinges on fuel-importing countries’ ability to credibly commit to unilaterally stabilizing the cli-

mate to compensate for producers' inaction; in reality, fossil-fuel exporters may doubt that importing countries are willing to make such sacrifices.

4 Conclusion

This paper presented a broad overview of the international climate policy regime and examined mechanisms through which climate ambition could be strengthened. I first demonstrated that the current regime fails to guarantee decarbonization in line with the Paris temperature target, as the sum of NDC targets exceeds the required emissions level. In this context, ITMOs will remain under-priced, and their unfettered use risks weakening overall ambition.

Then, I explored different options to regulate ITMOs. By restricting the use of ITMOs based on the adequacy of the seller's NDC, existing proposals (by e.g. ?) would limit hot air, but ITMOs would remain under-priced. To fully restore ambition and fairness, I proposed that a coalition of the willing submit a joint NDC aligned with the Paris target, and commit to exchanging ITMOs only among coalition members. I also suggested intermediary steps, such as strengthening reporting requirements to harmonize NDCs and establishing the principle that no ITMO should be sold by a country failing to meet its own NDC.

Finally, I analyzed alternative proposals to ratchet up ambition, such as differentiated carbon price floors, a fossil fuel non-proliferation treaty, and a refunding club. Alternatives appeared to either lack efficiency, lack ambition, or lack political realism, with the exception of JETPs. If ambitiously expanded in terms of scope and funding, these ongoing initiatives could boost both mitigation and climate finance (???)

The success of international climate policy will likely require both a tighter regulation of ITMOs and an expansion of JETPs. In both cases, a coalition of the willing seems the most viable option to reap the gains from the cooperation of ambitious countries.

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Supplementary Tables

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Beyond Current Initiatives:

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ITMO Regulations to Ratchet up Ambition

Table A1: Description of possible international policies to phase out fossil fuels.

International policy	Description
(<i>Status quo</i>) Unregulated ITMOs	Countries trade Internationally Transferred Mitigation Outcomes, bringing flexibility to the location of NDCs' emission reductions.
Partial linkage of carbon markets (?)	Carbon markets such as the EU ETS would accept external ETS allowance or emission reduction certificates up to some limit.
ITMOs + country-level integrity (?)	ITMOs with extra rules preventing countries lacking ambition to participate.
ITMOs avoiding ambition gap	ITMOs with extra rules (described in Section 2.4) ensuring that countries trading ITMOs have joint NDCs in line with the Paris target.
(<i>Status quo</i>) JETPs (?)	Just Energy Transition Partnerships where one developing country obtains concessional loans from a set of HICs to decarbonize its power sector.
JETPs with more grants (?)	JETPs financed by grants more than loans, of \$120 billion per year.
JETPs with wider scope (?)	JETPs with grants conditional on implementation of climate policy such as national carbon pricing.
Differentiated price floors (?)	Coordinated carbon price floors (\$25/tCO ₂ for LICs and lower-MICs, \$50 for upper-MICs, \$75 for HICs), with little revenue sharing between countries.
Carbon price on CBAM sectors (?)	International carbon price on carbon-intensive manufacturing sectors, with little revenue sharing between countries, either uniform price or differentiated price floors.
Climate club (???); Refunding club (??)	Uniform carbon price, with little revenue sharing between countries, with a CBAM, and dissuasive tariffs on imports from outside the club. Refunding clubs add initial payments and performance-based refunds to incentivize compliance.
Carbon price incentive (?)	Countries of a coalition receive (pay) transfers to the extent they price carbon above (below) a benchmark and emit less (more) than global average.
Fossil-Free Union (FFU) (?)	International cap-and-trade, with revenue returned on a basis given by an equal per capita benchmark with some adjustments.
FFU + Sustainable Union (SU) (?)	International cap-and-trade and new taxes (especially on wealth), where international transfers are proportional to the difference between a country's GNI per capita to the world average.
Uniform price floor + SU	Sustainable Union with a uniform carbon tax rather than a cap-and-trade.
Fossil non-proliferation treaty (??)	Coordinated phase out of fossil fuel extraction, with supply cuts starting in richest countries and ending with poorer, more fossil-dependent ones.
Producer carbon tax (?)	Uniform carbon tax applied on extraction or imports of fossil fuels, with part of the revenue shared with LICs and tariffs.
Expansion of climate finance (?????)	Reforms to the financial system to orient investment towards sustainable projects in the Global South, through public multilateral guarantees on climate projects, expansion of Multilateral Development Banks' (MDBs) operations, rechannelling of Special Drawing Rights to MDBs' capital, debt-for-climate swaps, money creation, etc.
Standards and bans	Implementation of common sectoral norms, e.g. standards on the CO ₂ -emission intensity of cars, shipping or aviation fuel; bans of fossil-fuel exploration, or on the opening of new coal power plants; common taxonomy for climate finance.

Table A2: Pros and cons of possible international policies.

International policy	Pros	Cons
(<i>Status quo</i>) Unregulated ITMOs	Cross-border financing of efficient decarbonization projects.	<i>Hot air</i> , risks weakening domestic climate action.
Partial linkage of carbon markets	Same as ITMOs.	Same as ITMOs.
ITMOs + country-level integrity	ITMOs with reduced hot air.	Either ambition gap or risks of unfair burden-sharing.
ITMOs avoiding ambition gap	ITMOs without ambition gap.	Trading between countries rather than firms, weakening enforcement.
(<i>Status quo</i>) JETPs	Cross-border financing of electricity decarbonization.	Limited scope; few grants; no effect on high emitting countries.
JETPs with more grants	JETPs with North–South transfers.	Limited scope; no effect on high emitting countries.
JETPs with wider scope	Potentially full country decarbonization.	No effect on high emitting countries.
Differentiated price floors	Country-wide efficiency; ambition adapted to country circumstances.	Few North–South transfer; no gains from trade.
Diff. prices on CBAM sectors	Decarbonization of manufacturing (efficient if uniform price).	Few North–South transfer (none if uniform price); limited scope.
Climate club; Re-funding club	Efficient decarbonization.	Few North–South transfer; trade sanctions may fail to incentivize recalcitrant countries and will hurt the club.
Carbon price incentive	Incentivizes decarbonization; respects sovereignty.	Weak enforcement capacity; wealthy countries may receive transfers.
Fossil-Free Union (FFU)	Efficient decarbonization with North–South transfers.	Ambition and burden-sharing rigid to changing circumstances.
FFU + Sustainable Union (SU)	Efficient decarbonization with large North–South transfers, spurring development.	Climate ambition rigid to changing circumstances; imperfect incentives for countries to implement complementary climate policies (as international transfers don't depend on the country's emissions).
Uniform price floor + SU	Efficient decarbonization with large North–South transfers, spurring development.	Climate ambition not guaranteed (price may be too low); imperfect incentives for countries to implement complementary climate policies.
Fossil non-proliferation treaty	Decarbonization.	Relies on the (unlikely) participation of fossil-fuel producing countries; would increase oil rents and hurt consumers, especially low-income ones; lacks efficiency.
Producer carbon tax	Efficient decarbonization.	Relies on the (unlikely) participation of fossil-fuel producing countries; would increase oil rents and hurt consumers, especially low-income ones.
Expansion of climate finance	Lower interest rates in LMICs, spurring sustainable development.	Does not cap emissions.
Standards and bans	Decarbonizes one sector.	Limited scope; no North–South transfer.

Table A3: Comparison summary of possible international policies.

International policy	Emission reductions	Least cost	Fair		Acceptable by			Oil countries	Flexible
			Rich pay	Poor gain	LICs	MICs	HICs		
(<i>Status quo</i>) Unregulated ITMOs	0	+	0	0	+++	+++	+++	+++	+++
Partial linkage of carbon markets	0	+	0	0	+++	+++	+++	+++	+++
ITMOs + country-level integrity	+	+	+	+	+	+	++	++	+++
ITMOs avoiding ambition gap	+++	+++	++	++	+++	++	+	--	-
(<i>Status quo</i>) JETPs	+	0	0	+	+++	+++	+++	+++	+++
JETPs with more grants	+	0	++	++	+++	++	+	--	+++
JETPs covering broad policy	++	+	++	++	+++	++	+	--	+++
Differentiated price floors	+	+	0	-	-	+	+++	-	+
Uniform price on CBAM sectors	++	++	0	--	-	+	+++	--	+
Diff. prices on CBAM sectors	+	+	+	-	0	++	++	--	+
Climate club; Refunding club	+++	+++	0	-	+	+	+++	---	+
Carbon price incentive	++	++	+	+	++	+	+	-	++
Fossil-Free Union (FFU)	++++	+++	++	++	+++	++	+	--	--
FFU + Sustainable Union (SU)	++++	+++	+++	+++	+++	++	0	---	--
Uniform price floor + SU	++	++	+++	+++	+++	+++	0	--	+
Fossil non-proliferation treaty	+	-	-	-	-	-	-	-	+
Producer carbon tax	++	++	--	---	---	--	-	-	+
Expansion of climate finance	++	+	+	+	+++	+++	++	+	+++
Standards and bans	++	0	0	0	+	+	++	+	0